

Oliver John (OJ) Watson

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EMPLOYMENT

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|---|----------------------------|
| Imperial College London | Oct 2023 – Present |
| <ul style="list-style-type: none">Associate Professor in Infectious Disease Modelling and Health Emergencies; Eric and Wendy Schmidt AI in Science Fellow | |
| London School of Hygiene and Tropical Medicine | Oct 2021 – Sep 2023 |
| <ul style="list-style-type: none">Schmidt Science Fellow | |
| Imperial College London | Apr 2020 – Sep 2021 |
| <ul style="list-style-type: none">Postdoctoral Researcher in infectious disease modelling | |
| Brown University | Oct 2019 – Mar 2020 |
| <ul style="list-style-type: none">Postdoctoral Researcher in computational genetics, bioinformatics | |

EDUCATION

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| Imperial College London | Oct 2015 – Sept 2019 |
| PhD | Wellcome Trust Funded 1+3 PhD Student (October 2016 – October 2019. Viva Date 18 Nov 2019): <ul style="list-style-type: none">Thesis Title: “Integrating genetic information into malaria transmission modelling”Supervisors: Prof Azra Ghani, Dr Lucy Okell and Dr Robert Verity |
| MRes | Biomedical Research (2015 – 2016): Dissertation <ul style="list-style-type: none">Dean’s Prize Awardee for top overall mark within MRes |
| University of Cambridge, Pembroke College | Oct 2011 – Jun 2015 |
| MSci | Systems Biology (2014 – 2015): 1st Class (Top bioinformatics research project mark) |
| BA | Natural Sciences (2011 – 2014): 1st Class (Foundation and College Scholarships for examination results) |

SCIENTIFIC PROFILE

Evidence of Esteem

- 78 authored articles and preprints (75 published; 3 preprints), plus 2 book chapters.
- Invited seminars, policy briefings, conference presentations and panel chairing.
- Undergraduate, Masters and PhD Supervision experience (Brown, Oxford) and Masters Examination (Oxford, Imperial)
- Track record of securing fellowship funding, contributing to international interdisciplinary grants (4 NIH grants, US CDC, CEPI, WHO) and securing grants as principal investigator (Wellcome).

Committee and Consortium Memberships

- WHO pfhrp23 and MFT Technical Panels (2023 - Present):** Technical and modelling input to WHO malaria diagnostic and treatment policy, including the 2024 pfhrp2 response plan and multiple first-line therapy guidance.
- ROpenSci (2018 - Present):** Community of software developers building open source and reproducible tools using the R programming language. I have contributed R packages and code reviews for data access and survey tools.
- VacSafeWG (2022 - 2024):** Schmidt Futures-funded vaccine safety working group in Africa. Contributed to policy briefs for supranational agencies on vaccines, data and early warning.

Academic Service

- Journal Reviewer:** Nature, PNAS, Nature Communications, The Lancet, PLoS Medicine and other journals. Full list under Professional Membership and Service.
- Grant Reviewer:** Wellcome, NIH, Horizon Europe, Netherlands Organisation for Health Research and Development, Imperial College London (Internal Fellowship Rounds)

RESEARCH SUPPORT

Grants and Fellowships

As Principal Investigator

The UK Academy of Medical Sciences Springboard. Dr Oliver Watson (PI). Deep Learning Surrogates of Infectious Disease Models. £125,000.00.	2025 - 2027
MRC GIDA Pump Priming Funding. Dr Oliver Watson (PI). Hackathon - Bridging the Gap: Applied Infectious Disease Epidemiology Tools to Support Evidence-Informed Decision-Making. £31,151.49.	2024 - 2026
Community Jameel. Dr Oliver Watson (PI). Jameel Institute Real-time Intelligent Support for Emergencies (RISE): Vaccine and Mortality Modelling in the Gaza Conflict. £1,400,000 total programme: £700,000 to Imperial and £700,000 to LSHTM and Geneva partners.	2024 - 2026
Wellcome Trust. Dr Oliver Watson (PI). Leveraging forecasts to optimise decision making. £79,365	2024 - 2025
Imperial College Research Fellowship supported by an Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship. Dr Oliver Watson (PI). Novel methods and data sources for pandemic preparedness. £337,465	2023 - 2027
Schmidt Science Fellowship. Dr Oliver Watson (PI). Vaccine impact assessments and mortality estimation from COVID-19. \$200,000.	2021 - 2023
Wellcome Trust. Mr Oliver Watson (PhD Funding). Integration of parasite genetic information in malaria transmission modelling. £165,784.00	2015 - 2019

As Co-Investigator

NIH, 1R01AI190302-01. Prof Jonathan Juliano (PI). Predicting the spread of antimalarial drug resistance using deep learning surrogates. Named co-investigator. Approximately £650,000 to Imperial College London; Imperial subaward PI Robert Verity. October 2025 - September 2030.	2025 - 2030
MRC Mid Equipment Grant. Dr Rich FitzJohn (PI). Accelerating Infectious Disease Modelling Using General Purpose GPUs. £186,400.00. Named co-investigator.	2025 - 2026
Coalition for Epidemic Preparedness Innovations. Prof Azra Ghani (PI). Assessing and predicting the public health benefits and economic health impact of CEPI's investments for COVID-19. £32,000 to Imperial College London.	2023
US CDC, U01GH002319. Prof Francesco Checchi (PI). Covid-19 and related public health threats in populations affected by crises: a multi-disciplinary, collaborative research programme. \$15,000,000 (Initial Award. Annual funder obligations reviewed through cooperative agreement). Named co-investigator, contribution in kind to ensure LMIC partner funding in response to funding reductions	2022 - 2027
NIH, 1R01AI177791-01A1. Dr Jonathan Parr (PI). Epidemiology and determinants of emerging artemisinin resistant malaria in Ethiopia. \$3,603,465. Named co-investigator. \$42,000 over 3 years.	2023 - 2028
NIH, 1R01AI156267-01A1. Dr Jeffrey Bailey (PI). Artemisinin Resistance in Africa: its emergence and evolution in Rwanda. \$3,989,037. Named co-investigator. \$60,000 over 3 years	2021 - 2026
NIH, 1R03AI173899-01A1. Dr Jessica Lin (PI). Does treating low density malaria infections reduce malaria transmission? \$100,000. Named co-investigator. \$15,000 over 2 years	2023 - 2025
World Health Organisation. Dr Alexandra Hogan (PI). Modelling the Impact of COVID-19 Vaccination Strategies. \$100,533. Drafted submission. \$52,325 allocated for my 100% FTE on grant as named researcher.	2021
Wellcome Trust. Dr Patrick Walker (PI). Modelling the global spread and determinants of COVID-19 transmission. £96,664. Drafted submission. £30,255 allocated for 100% FTE on grant as named researcher.	2020

Invited Consulting

Clinton Health Access Initiative. Modelling the impact of alternative RDTs in sub-Saharan Africa	2024-2025
Clinton Health Access Initiative. Modelling the spread of antimalarial resistance in sub-Saharan Africa.	2023 - 2024
World Health Organization. Modelling the global risk and spread of pfhrp2/3 gene deletions.	2022 - 2023
Asian Development Bank. Evaluating COVID-19 vaccine impact and long term vaccination strategies in India.	2021 - 2022

PUBLICATIONS

* = Equal Author Contribution

1. Ghafari M, **Watson OJ**, Dascalu S, Mon SHH, Abeler-Dörner L, Rezaee-Zavareh MS, Pley C, Kordasti S, Abbas K, Madani K, Panovska-Griffiths J. Global vaccine readiness: equity-by-design in pandemic preparedness and response. *Expert Rev. Vaccines*. 25, 2732508 (2026). doi:10.1080/14760584.2026.2732508
2. Wernsman Young N, Meier-Scherling CPG, Cuomo-Dannenburg G, Tollefson GA, Connelly SV, Marglous J, Gerdes Gyuricza I, Carey-Ewend K, Kyong-Shin R, Popkin-Hall ZR, Zeleke AJ, Ishengoma DS, Fola AA, Simkin A, Niaré K, Parr JB, Conrad M, Okell LC, Ruybal-Pesántez S, **Watson OJ**, Juliano JJ, Bailey JA, Verity R. Mapping the prevalence of molecular markers of *Plasmodium falciparum* artemisinin partial resistance in Africa: a systematic review and spatiotemporal modelling study. *Lancet Infect. Dis.* (2026). doi:10.1016/S1473-3099(26)00237-9
3. Christen P, Dawa J, Ahmed O, Akinyi F, Masakhwe BA, Bekalo DB, Biwot C, Bosire AB, Charles G, Gathoni V, Gaythorpe KAM, Geoffrey G, Gowa S, Hamumy FS, Kagiri H, Kamundia G, Kamunya S, Kariuki P, Khalayi L, Kombe I, Kwamboka P, Liétar P, Likalamu N, Lugonzo B, Mariita V, Masinde A, Mbithi M, McCabe R, Monari FN, Motiri FO, Muchiri AM, Mudamba A, Mugo PW, Mutono N, Muriithi M, Musasia R, Mutunga M, Mwangi W, Nalwa J, Ngeno KI, Nyaberi JM, Nzoka PM, Ojal J, Okech T, Okunga E, Omondi E, Osoro J, Osoro P, Patel A, Sayianka S, Shigoli T, Silali C, Thendu MM, van Elsland SL, Wanjiku KA, Muchemi P, Whittles LK, Winskill P, Ombajo LA, Thumbi SM, **Watson OJ**. Bridging the gap between public health, academia and policy. *BMJ Glob. Health*. 11, e019587 (2026). doi:10.1136/bmjgh-2025-019587
4. Christen P, Ombajo LA, Cori A, Dawa J, Djaafara BA, Kimani TN, Schneider CMJ, van Elsland SL, Thumbi M, Veras M, Whittaker C, Whittles LK, **Watson OJ**. Enhancing epidemic forecast usability for policymakers: A global mixed-methods study. *PLOS Glob. Public Health*. 6, e0006519 (2026). doi:10.1371/journal.pgph.0006519
5. Gerdes Gyuricza I, Fola AA, Simkin A, Thwai KL, Juliano JJ, Bailey JA, Johri P, Henry CM, Cabrera-Sosa L, Porras-Laymito G, Cheng Q, **Watson OJ**, Gamboa D, Valdivia HO, Parr JB. Malaria control and the unexpected spread of diagnostic-resistant *Plasmodium falciparum* in Peru. *bioRxiv* (2026). doi:10.64898/2026.02.25.707493
6. Boni MF, Soulama I, Opigo J, **Watson OJ**, Ogutu B. Slowing artemisinin resistance in Africa. *Sci. Adv.* 11, eaeb7618 (2025). doi:10.1126/sciadv.aeb7618
7. Benjamin H, Barnsley G, **Watson OJ**, Jit M. The global return-on-investment of COVID-19 vaccines in the first year of the vaccination programme. *medRxiv* (2025). doi:10.1101/2025.09.02.25334932
8. Juliano JJ, Meier-Scherling CPG, Wernsman Young N, Tollefson GA, Connelly SV, Parr JB, Conrad MD, Sadler JM, Hennelly CM, Assefa A, Okell L, Fola AA, Niaré K, Marglous J, Carey-Ewend K, Futala Kyong-Shin R, Gerdes Gyuricza I, Cuomo-Dannenburg G, Ruybal-Pesántez S, **Watson OJ**, Verity R, Bailey JA. Toward setting minimum and optimal data to report for malaria molecular surveillance with targeted sequencing: The “what” and “why.” *Am. J. Trop. Med. Hyg.* 113, 1169–1177 (2025). doi:10.4269/ajtmh.25-0147
9. Kraemer MUG, Tsui JL-H, Chang SY, Lytras S, Khurana MP, Vanderslott S, Bajaj S, Scheidwasser N, Curran-Sebastian JL, Semenova E, Zhang M, Unwin HJT, **Watson OJ**, Mills C, Dasgupta A, Ferretti L, Scarpino SV, Koua E, Morgan O, Tegally H, Paquet U, Moutsianas L, Fraser C, Ferguson NM, Topol EJ, Duchêne DA, Stadler T, Kingori P, Parker MJ, Dominici F, Shadbolt N, Suchard MA, Ratmann O, Flaxman S, Holmes EC, Gomez-Rodriguez M, Schölkopf B, Donnelly CA, Pybus OG, Cauchemez S, Bhatt S. Artificial intelligence for modelling infectious disease epidemics. *Nature*. 638, 623–635 (2025)
10. Wernsman Young N, Gashema P, Giesbrecht D, Munyaneza T, Maisha F, Mwebembezi F, Budodo R, Leonetti A, Crudale R, Iradukunda V, Jean Bosco N, Kirby RI, Boyce RM, Mandara CI, Kanyankole GK, Ntaro M, Okell LC, **Watson OJ**, Mulogo E, Ishengoma DS, Hangi S, Karema C, Mazarati J-B, Juliano JJ, Bailey JA. High frequency of artemisinin partial resistance mutations in the great lakes region revealed through rapid pooled deep sequencing. *J. Infect. Dis.* 231, 269–280 (2025)
11. Whittaker C, Barnsley G, Olivera Mesa D, Cox V, Laydon DJ, Tan CW, Zhu F, Johnson R, Doohan P, Whittles LK, Nedjati-Gilani G, Winskill P, Hogan AB, Deol A, Mukandavire C, Hauck K, Lye DCB, Wang L-F, **Watson OJ**, Ghani AC. Quantifying the impact of a broadly protective sarbecovirus vaccine in a future SARS-X pandemic. *Nat. Commun.* 16, 8495 (2025). doi:10.1038/s41467-025-63399-x
12. **Watson OJ**, Muchiri S, Ward A, Meier-Sherling C, Asua V, Katairo T, Brewer T, Cuomo-Dannenburg G, Winskill P, Bailey JA, Okell L, Scudu G, Woolsey AM. Risk of selection and timelines for the continued spread of artemisinin and partner drug resistance in Africa. *medRxiv* (2024), doi:10.1101/2024.08.28.24312699. <http://hdl.handle.net/10044/1/114684>
13. **Watson OJ**, Hogan AB. Impact of COVID-19 vaccination programmes in Europe: lives saved and lessons learned. *Lancet Respir. Med.* 12, 663–664 (2024)
14. Meier-Scherling CPG, **Watson OJ**, Asua V, Ghinai I, Katairo T, Garg S, Conrad MD, Rosenthal PJ, Okell LC, Bailey JA. Selection of *Plasmodium falciparum* kelch13 mutations in Uganda in comparison with southeast Asia: a modelling study. *Lancet Microbe*. 6, 101027 (2025)
15. **Watson OJ**, Tran TN-A, Zupko RJ, Symons T, Thomson R, Visser T, Rumisha S, Dzianach PA, Hathaway N, Kim I, Juliano JJ, Bailey JA, Slater H, Okell L, Gething P, Ghani A, Boni MF, Parr JB, Cunningham J. Global risk of selection and spread of *Plasmodium falciparum* histidine-rich protein 2 and 3 gene deletions. *Nature Medicine*. 2025;31:3372-3379. doi:10.1038/s41591-025-03974-3

16. Hogan AB, Wu SL, Toor J, Olivera Mesa D, Doohan P, **Watson OJ**, Winskill P, Charles G, Barnsley G, Riley EM, Khoury DS, Ferguson NM, Ghani AC. Long-term vaccination strategies to mitigate the impact of SARS-CoV-2 transmission: A modelling study. *PLoS Med.* **20**, e1004195 (2023)
17. Barnsley G, Olivera Mesa D, Hogan AB, Winskill P, Torkelson AA, Walker DG, Ghani AC, **Watson OJ**. Impact of the 100 days mission for vaccines on COVID-19: a mathematical modelling study. *Lancet Glob. Health.* **12**, e1764–e1774 (2024)
18. Bhatia S*, Imai N*, **Watson OJ***, Abbood A, Abdelmalik P, Cornelissen T, Ghazzi S, Lassmann B, Nagesh R, Ragonnet-Cronin ML, Schnitzler JC, Kraemer MU, Cauchemez S, Nouvellet P, Cori A. Lessons from COVID-19 for rescalable data collection. *Lancet Infect. Dis.* **23**, e383–e388 (2023)
19. Hogan AB, Doohan P, Wu SL, Mesa DO, Toor J, **Watson OJ**, Winskill P, Charles G, Barnsley G, Riley EM, Khoury DS, Ferguson NM, Ghani AC. Estimating long-term vaccine effectiveness against SARS-CoV-2 variants: a model-based approach. *Nat. Commun.* **14**, 4325 (2023)
20. Chemwor GC, Andagalu BM, Onyango IA, Opot BH, Okoth RO, Yedah RA, Juma JA, Mwakio EW, Wakoli DM, Amwoma JG, Cheruiyot AC, Juma DW, Ogutu BR, Egbo TE, Garges EC, Roth AL, Kamau E, **Watson OJ**, Akala HM. Therapeutic response to artemisinin combination therapies among individuals with *Plasmodium falciparum* single infection vs mixed *Plasmodium* species infections: a retrospective posthoc analysis in Kisumu County, western Kenya. *Int. J. Infect. Dis.* **132**, 17–25 (2023)
21. Sheppard RJ, **Watson OJ**, Pieciak R, Lungu J, Kwenda G, Moyo C, Chanda SL, Barnsley G, Brazeau NF, Gerard-Ursin ICG, Olivera Mesa D, Whittaker C, Gregson S, Okell LC, Ghani AC, MacLeod WB, Del Fava E, Melegaro A, Hines JZ, Mulenga LB, Walker PGT, Mwananyanda L, Gill CJ. Using mortuary and burial data to place COVID-19 in Lusaka, Zambia within a global context. *Nat. Commun.* **14**, 3840 (2023)
22. McCabe R, Whittaker C, Sheppard RJ, Abdelmagid N, Ahmed A, Alabdeen IZ, Brazeau NF, Ahmed Abd Elhameed AE, Bin-Ghouth AS, Hamlet A, AbuKoura R, Barnsley G, Hay JA, Alhaffar M, Koum Besson E, Saje SM, Sisay BG, Gebreyesus SH, Sikamo AP, Worku A, Ahmed YS, Mariam DH, Sisay MM, Checchi F, Dahab M, Endris BS, Ghani AC, Walker PGT, Donnelly CA, **Watson OJ**. Alternative epidemic indicators for COVID-19 in three settings with incomplete death registration systems. *Sci Adv.* **9**, eadg7676 (2023)
23. Paschalidis A*, **Watson OJ***, Aydemir O, Verity R, Bailey JA. coiaf: Directly estimating complexity of infection with allele frequencies. *PLoS Comput. Biol.* **19**, e1010247 (2023)
24. Kirby R, Giesbrecht D, Karema C, **Watson O**, Lewis S, Munyaneza T, Butera JDD, Juliano JJ, Bailey JA, Mazarati J-B. Examining the Early Distribution of the Artemisinin-Resistant *Plasmodium falciparum* kelch13 R561H Mutation in Areas of Higher Transmission in Rwanda. *Open Forum Infect Dis.* **10**, ofad149 (2023)
25. Andagalu B, **Watson OJ**, Onyango I, Opot B, Okoth R, Chemwor G, Sifuna P, Juma D, Cheruiyot A, Yeda R, Okudo C, Wafubwa J, Yalwala S, Abuom D, Ogutu B, Cowden J, Akala HM, Kamau E. Malaria Transmission Dynamics in a High-Transmission Setting of Western Kenya and the Inadequate Treatment Response to Artemether-Lumefantrine in an Asymptomatic Population. *Clin. Infect. Dis.* **76**, 704–712 (2023)
26. Andagalu B, **Watson OJ**, Onyango I, Opot B, Okoth R, Chemwor G, Sifuna P, Juma D, Cheruiyot A, Yeda R, Okudo C, Wafubwa J, Yalwala S, Abuom D, Ogutu B, Cowden J, Akala HM, Kamau E. Reply to Blanken et al. *Clin. Infect. Dis.* **76**, 557–559 (2023)
27. **Watson OJ***, Barnsley G*, Toor J, Hogan AB, Winskill P, Ghani AC. Global impact of the first year of COVID-19 vaccination: a mathematical modelling study. *Lancet Infect. Dis.* **22**, 1293–1302 (2022)
28. **Watson OJ***, Gao B*, Nguyen TD*, Tran TN-A, Penny MA, Smith DL, Okell L, Aguas R, Boni MF. Pre-existing partner-drug resistance to artemisinin combination therapies facilitates the emergence and spread of artemisinin resistance: a consensus modelling study. *Lancet Microbe.* **3**, e701–e710 (2022)
29. Whittaker C, **Watson OJ**, Alvarez-Moreno C, Angkasekwina N, Boonyasiri A, Carlos Triana L, Chanda D, Charoenpong L, Chayakulkeeree M, Cooke GS, Croda J, Cucunubá ZM, Djaafara BA, Estofotele CF, Grillet ME, Faria NR, Figueiredo Costa S, Forero-Peña DA, Gibb DM, Gordon AC, Hamers RL, Hamlet A, Irawany V, Jitmuang A, Keurueangkul N, Kimani TN, Lampo M, Levin AS, Lopardo G, Mustafa R, Nayagam S, Ngamprasertchai T, Njeri NIH, Nogueira ML, Ortiz-Prado E, Perroud MW, Phillips AN, Promsin P, Qavi A, Rodger AJ, Sabino EC, Sangkaew S, Sari D, Sirijatuphat R, Sposito AC, Srisangthong P, Thompson HA, Udwadia Z, Valderrama-Beltrán S, Winskill P, Ghani AC, Walker PGT, Hallett TB. Understanding the Potential Impact of Different Drug Properties on Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2) Transmission and Disease Burden: A Modelling Analysis. *Clin. Infect. Dis.* **75**, e224–e233 (2022)
30. Ghafari M*, **Watson OJ***, Karlinsky A, Ferretti L, Katzourakis A. A framework for reconstructing SARS-CoV-2 transmission dynamics using excess mortality data. *Nat. Commun.* **13**, 3015 (2022)
31. Brazeau NF, Verity R, Jenks S, Fu H, Whittaker C, Winskill P, Dorigatti I, Walker PGT, Riley S, Schnekenberg RP, Hoeltgebaum H, Mellan TA, Mishra S, Unwin HJT, **Watson OJ**, Cucunubá ZM, Baguelin M, Whittles L, Bhatt S, Ghani AC, Ferguson NM, Okell LC. Estimating the COVID-19 infection fatality ratio accounting for seroreversion using statistical modelling. *Commun. Med.* **2**, 54 (2022)
32. Pons-Salort M, John J, **Watson OJ**, Brazeau NF, Verity R, Kang G, Grassly NC. Reassessing Reported Deaths and Estimated Infection Attack Rate during the First 6 Months of the COVID-19 Epidemic, Delhi, India. *Emerg. Infect. Dis.* **28**, 759–766 (2022)

33. Olivera Mesa D, Hogan AB, **Watson OJ**, Charles GD, Hauck K, Ghani AC, Winskill P. Modelling the impact of vaccine hesitancy in prolonging the need for Non-Pharmaceutical Interventions to control the COVID-19 pandemic. *Commun. Med.* 2, 14 (2022)
34. Favas C, Jarrett P, Ratnayake R, **Watson OJ**, Checchi F. Country differences in transmissibility, age distribution and case-fatality of SARS-CoV-2: a global ecological analysis. *Int. J. Infect. Dis.* 114, 210–218 (2022)
35. McCabe R, Kont MD, Schmit N, Whittaker C, Løchen A, Walker PGT, Ghani AC, Ferguson NM, White PJ, Donnelly CA, **Watson OJ**. Communicating uncertainty in epidemic models. *Epidemics*. 37, 100520 (2021)
36. Mousa A, Winskill P, **Watson OJ**, Ratmann O, Monod M, Ajelli M, Diallo A, Dodd PJ, Grijalva CG, Kiti MC, Krishnan A, Kumar R, Kumar S, Kwok KO, Lanata CF, de Waroux OLP, Leung K, Mahikul W, Melegaro A, Morrow CD, Mossong J, Neal EF, Nokes DJ, Pan-Ngum W, Potter GE, Russell FM, Saha S, Sugimoto JD, Wei WI, Wood RR, Wu J, Zhang J, Walker P, Whittaker C. Social contact patterns and implications for infectious disease transmission - a systematic review and meta-analysis of contact surveys. *Elife*. 10, e70294 (2021)
37. Whittaker C, Walker PGT, Alhaffar M, Hamlet A, Djaafara BA, Ghani A, Ferguson N, Dahab M, Checchi F, **Watson OJ**. Under-reporting of deaths limits our understanding of true burden of covid-19. *BMJ*. 375, n2239 (2021)
38. Mangal T, Whittaker C, Nkhoma D, Ng'ambi W, Watson O, Walker P, Ghani A, Revill P, Colbourn T, Phillips A, Hallett T, Mfutso-Bengo J. Potential impact of intervention strategies on COVID-19 transmission in Malawi: a mathematical modelling study. *BMJ Open*. 11, e045196 (2021)
39. Imai N, Hogan AB, Williams L, Cori A, Mangal TD, Winskill P, Whittles LK, **Watson OJ**, Knock ES, Baguelin M, Perez-Guzman PN, Gaythorpe KAM, Sonabend R, Ghani AC, Ferguson NM. Interpreting estimates of coronavirus disease 2019 (COVID-19) vaccine efficacy and effectiveness to inform simulation studies of vaccine impact: a systematic review. *Wellcome Open Res.* 6, 185 (2021)
40. Knock ES, Whittles LK, Lees JA, Perez-Guzman PN, Verity R, FitzJohn RG, Gaythorpe KAM, Imai N, Hinsley W, Okell LC, Rosello A, Kantas N, Walters CE, Bhatia S, **Watson OJ**, Whittaker C, Cattarino L, Boonyasiri A, Djaafara BA, Fraser K, Fu H, Wang H, Xi X, Donnelly CA, Jauneikaite E, Laydon DJ, White PJ, Ghani AC, Ferguson NM, Cori A, Baguelin M. Key epidemiological drivers and impact of interventions in the 2020 SARS-CoV-2 epidemic in England. *Sci. Transl. Med.* 13 (2021), doi:10.1126/scitranslmed.abg4262
41. McCabe R, Kont MD, Schmit N, Whittaker C, Løchen A, Baguelin M, Knock E, Whittles LK, Lees J, Brazeau NF, Walker PG, Ghani AC, Ferguson NM, White PJ, Donnelly CA, Hauck K, **Watson OJ**. Modelling intensive care unit capacity under different epidemiological scenarios of the COVID-19 pandemic in three Western European countries. *Int. J. Epidemiol.* 50, 753–767 (2021)
42. Brazeau NF, Mitchell CL, Morgan AP, Deutsch-Feldman M, **Watson OJ**, Thwai KL, Gelabert P, van Dorp L, Keeler CY, Waltmann A, Emch M, Gartner V, Redelings B, Wray GA, Mwandagaliwa MK, Tshetu AK, Likwela JL, Edwards JK, Verity R, Parr JB, Meshnick SR, Juliano JJ. The epidemiology of *Plasmodium vivax* among adults in the Democratic Republic of the Congo. *Nat. Commun.* 12, 4169 (2021)
43. Barnett-Howell Z, **Watson OJ**, Mobarak AM. The benefits and costs of social distancing in high- and low-income countries. *Trans. R. Soc. Trop. Med. Hyg.* 115, 807–819 (2021)
44. Smith TP, Flaxman S, Gallinat AS, Kinoshian SP, Stemkovski M, Unwin HJT, **Watson OJ**, Whittaker C, Cattarino L, Dorigatti I, Tristem M, Pearce WD. Temperature and population density influence SARS-CoV-2 transmission in the absence of nonpharmaceutical interventions. *Proc. Natl. Acad. Sci. U. S. A.* 118 (2021), doi:10.1073/pnas.2019284118
45. Djaafara BA, Whittaker C, **Watson OJ**, Verity R, Brazeau NF, Widyastuti, Oktavia D, Adrian V, Salama N, Bhatia S, Nouvellet P, Sherrard-Smith E, Churcher TS, Surendra H, Lina RN, Ekawati LL, Lestari KD, Andrianto A, Thwaites G, Baird JK, Ghani AC, Elyazar IRF, Walker PGT. Using syndromic measures of mortality to capture the dynamics of COVID-19 in Java, Indonesia, in the context of vaccination rollout. *BMC Med.* 19, 146 (2021)
46. Hogan AB, Winskill P, **Watson OJ**, Walker PGT, Whittaker C, Baguelin M, Brazeau NF, Charles GD, Gaythorpe KAM, Hamlet A, Knock E, Laydon DJ, Lees JA, Løchen A, Verity R, Whittles LK, Muhib F, Hauck K, Ferguson NM, Ghani AC. Within-country age-based prioritisation, global allocation, and public health impact of a vaccine against SARS-CoV-2: A mathematical modelling analysis. *Vaccine*. 39, 2995–3006 (2021)
47. **Watson OJ**, Alhaffar M, Mehchy Z, Whittaker C, Akil Z, Brazeau NF, Cuomo-Dannenburg G, Hamlet A, Thompson HA, Baguelin M, FitzJohn RG, Knock E, Lees JA, Whittles LK, Mellan T, Winskill P, Imperial College COVID-19 Response Team, Howard N, Clapham H, Checchi F, Ferguson N, Ghani A, Beals E, Walker P. Leveraging community mortality indicators to infer COVID-19 mortality and transmission dynamics in Damascus, Syria. *Nat. Commun.* 12, 2394 (2021)
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78. **Watson OJ**, Slater HC, Verity R, Parr JB, Mwandagalirwa MK, Tshefu A, Meshnick SR, Ghani AC. Modelling the drivers of the spread of Plasmodium falciparum hrp2 gene deletions in sub-Saharan Africa. Elife. 6, e25008 (2017)

Book Chapters

- Routledge I, **Watson OJ**, Griffin JT, Ghani AC. "Predictive Malaria Epidemiology, Models of Malaria Transmission and Elimination" in Encyclopedia of Malaria, P. G. Kremsner, S. Krishna, Eds. (Springer New York, New York, NY, 2018). pp. 1–7
- Watson OJ**, Routledge I, Griffin JT, Ghani AC. "Predictive Malaria Epidemiology, Models of Malaria Control Interventions and Elimination" in Encyclopedia of Malaria, P. G. Kremsner, S. Krishna, Eds. (Springer New York, New York, NY, 2018). pp. 1–7

REPORTS

What models are available for predicting infectious outbreaks in Africa?. Chapter Contribution for Signature Initiative to Mitigate Biological Threats in Africa. A partnership initiative between the Africa CDC and the G7 led Global Partnership

Against the Spread of Weapons and Materials of Mass Destruction. Under Africa CDC Approval

Response plan to pfhpr2 gene deletions, second edition. Technical contributor to the World Health Organization response plan. Published 6 December 2024.

Imperial College COVID-19 Reports. Author of 19 Imperial College COVID-19 Reports including (Co)-First/Last Author of 6 Reports, all of which have led to peer-reviewed scientific publications. 2020-2021

UK COVID-19 Public Inquiry. First Witness Statement of Professor Neil Ferguson. Contributed writing to three sections as part of UK COVID-19 Public Inquiry. July 2023

A preliminary assessment of COVAX's impact in lower-income countries. Contributed analysis to GAVI white paper about the impact of the COVAX Facility on lives saved due to COVID-19 vaccination in LMICs. May 2023

From insight to action: Examining mortality in Somalia. Contributing author to WHO and Unicef report on Excess Mortality in Somalia due to drought. March 2023

Mortality patterns in Somalia: Retrospective estimates and scenario-based forecasting. Lead author of modelling analysis underpinning WHO and Unicef Policy Report on Mortality in Somalia. March 2023

Evaluating COVID-19 vaccine impact and long term vaccination strategies in India. Lead Author of Asian Development Bank Internal Report. March 2022

COVID-19: Preparing for the future. Looking ahead to winter 2021/22 and beyond. Contributed analysis to UK Academy of Medical Sciences Rapid Review. June 2021

An assessment of COVID-19 mortality in humanitarian contexts. Contributing Author to UNHCR COVID-19 Research. March 2021

SEMINARS AND INVITED PRESENTATIONS

Invited seminar. WHO Hub for Pandemic and Epidemic Intelligence, Berlin. 2025

Jameel Institute Real-time Intelligent Support for Emergencies (JI-RISE). Oral Presentation. Jameel Institute Annual Symposium. 20 November 2024

Mathematical modelling - a tool in evaluating drug policies and antimalarial resistance containment . Oral Presentation. WHO Technical consultation on multiple first-line therapies as part of the strategy to respond to antimalarial drug resistance. 15 May 2024

Timelines for the continued spread of artemisinin and partner drug resistance in Africa . Oral Presentation. WHO Steering Committee Meeting on Antimalarial Resistance. 9 April 2024

Experience of being a Schmidt Science Fellow. Panel Member. Rhodes Trust Trustee Meeting. Q&A with fundees panel event. 8 March 2024

Equitably equipping the response to conflicts and pandemics. Oral Presentation. Schmidt Science Fellows Alumni Conference, Toronto. 18 July 2024

How to use ChatGPT for writing modelling papers. Oral Presentation. Imperial College London. Seminar Training Series. 20 June 2024

Modeling to understand the past and predict the future. Oral Presentation. INES Runhegeri, Rwanda. 29 April 2024

International Advances in COVID-19. Framework for Scientific Evolution and Evaluation. Oral Presentation. UCSF Fresno Town Hall Lecture Series. 7 December 2023

Surrogate modelling of agent based model features: an application for evaluating the risk of malaria parasites evolving to evade diagnosis. Seminar. University of Bergen. Workshop on teaching and research in statistics and data science. 28 November 2023

Approaches to Navigating Science Beyond 2050. Chair of Panel Session. Schmidt Science Fellows 5 year Alumni Conference. 18 November 2023

Update on the state of COVID-19 underreporting in Africa. Afternoon. The impact of pfhpr2/3 deletions on future testing of malaria in Africa. Seminar. Johns Hopkins University. 21 July 2023

Global risk of selection and spread of Plasmodium falciparum histidine-rich protein 2 and 3 gene deletions: Implications for future RDT procurement and WHO malaria control policy guidance. Oral Presentation. President's Malaria Initiative RDT Task Force. 10 July 2023

Mortality patterns in Somalia: retrospective estimates and scenario-based forecasts. Oral Presentation. Somalia Ministry of Health; WHO EMRO; Horn of Africa Food Security and Nutrition Working Group. 30 March 2023

Risk based approach to transitioning to alternative RDTs. Oral Presentation. WHO Meeting to update response plan to pfhrp2/3 gene deletions. 26 January 2023

Pros and Cons of Global Roll-Out. Oral Presentation. International Workshop on COVID-19 Vaccines 2022 - December Edition organised by Virology Education. 15 December 2022

Integrating novel data sources for COVID-19 outbreak modelling. Seminar. Afya Research Consortium. Nairobi. 20 July 2022

COVID-19 Vaccines: Unfinished Business. Chair of Panel Session. International Symposium on Vaccines and Global Health. 1 April 2022

Lessons in persistence for fellowship rejections and practising equanimity in academia. Oral Presentation. Johns Hopkins University Lunchtime Seminar. 29 March 2022

State of the Pandemic. Oral Presentation. International Monetary Fund. 27 April 2021

Utilising vaccine safety and confidence data in support of modelling global vaccine roll out. Oral Presentation. Inaugural meeting of Columbia-Wits Vaccine Safety and Confidence-Building Working Group. 29 April 2021

Imperial SPI-M & covidsim.org Modelling Overview. Seminar. UK Cabinet Office. 16 March 2021

Using Earth Observation to understand the COVID-19 pandemic. Oral Presentation. Imperial Earth Observation Network. 25 February 2021

Impact of COVID-19 vaccination in India: a subnational mathematical modelling study. Oral Presentation. Asian Development Bank. 20 January 2021

Modelling COVID-19 epidemic in Khartoum. Oral Presentation. Sudan Federal Ministry of Health. 9 December 2020

COVID-19 Mortality Underreporting: Investigation into Damascus, Syria. Oral Presentation. Save The Children COVID-19. 16 November 2020

COVID-19 Mortality Underreporting: Investigations in Damascus, Khartoum and Aden. Oral Presentation. WHO COVID-19 Modeling Group. 23 October 2020

Modelling to support COVID-19 decision making. Oral Presentation. UK DfID Country Offices. 25 June 2020

Short-term forecasts of healthcare demand for countries in the early stage of COVID-19 epidemics. Oral Presentation. WHO COVID-19 Modeling Group. 15 May 2020

Infectious disease modelling 101- An introduction to COVID-19 models. Seminar. PATH. 14 April 2020

Integration of malaria parasite genetics into malaria transmission modelling. Seminar. Warren Alpert Medical School of Brown University. 22 October 2019

Modelling the epidemiological implications and explanations of discordances between microscopy and rapid diagnostic tests. Seminar. University of North Carolina at Chapel Hill. 1 November 2017

CONFERENCE AND MEETING PRESENTATIONS

Symposium organisation; led successful submission. JI-RISE team. Bridging academia and humanitarian response: Data, modelling and policy for crisis-affected populations. ASTMH Annual Meeting, Toronto. 11 November 2025. Session led by Bhargavi Rao.

Co-authored oral presentation; presented by Arran Hamlet. Hamlet A, ... Watson OJ. Jameel Institute Crisis Vaccination Planner: Prototype Modelling Tool to Prioritise Vaccination in Humanitarian Emergencies. MSF Scientific Day, London. 2025

Symposium Chair and Oral Presentation. **Watson OJ** et al. Artemisinin Resistance Response in Africa: Integrating Molecular Surveillance and Mathematical Modeling to Mitigate Emerging Risk and Impact on Malaria Burden Reduction. ASTMH. 2024

Oral Presentation. **Watson OJ** et al. Risk of selection and timelines for the continued spread of artemisinin and partner drug resistance in Africa. Multilateral Initiative On Malaria (MIM Society) 8th Pan-African Malaria Conference. 2024

Symposium Presentation. **Watson OJ** et al. Global risk of selection and spread of Plasmodium falciparum histidine-rich protein 2 and 3 gene deletions: Implications for future RDT procurement and malaria control policy guidance. Multilateral Initiative On Malaria (MIM Society) 8th Pan-African Malaria Conference. 2024

Poster Presentation. Barnsley G, ..., **Watson OJ**. Modelling the potential impact of 100 days mission and broader investments on the COVID-19 pandemic. 17th Vaccine Congress. 2023

Poster Presentation. Meier-Scherling C, **Watson OJ** et al. Comparison of strength of selection for Plasmodium falciparum artemisinin resistance-associated mutations between southeast Asia and Uganda. ASTMH. 2023

Poster Presentation. McCabe R, ..., **Watson OJ**. Alternative epidemic indicators for COVID-19 in three settings with incomplete death registration systems. ASTMH. 2023

Oral Presentation. Barnsley G, ..., **Watson OJ**. Modelling the potential impact of 100 Days Mission and broader investments on the COVID-19 pandemic. Epidemics 9. 2023

Invited Meeting Participant. Rhodes Policy Summit: Creating a Positive Legacy from the Pandemic. London. 2023

Invited Chair of Panel Session. Karlinsky A, **Watson OJ** et al. The old-new science of excess mortality. Royal Statistical Science Conference. 2023

Oral Presentation. **Watson OJ** et al. Vaccine Impact Estimation: Pros and Cons of the Global COVID-19 Vaccination Campaign and evaluation of improved R&D in the Republic of Korea. UK-Korea bilateral international meeting organised by The Royal Society. 2023

Poster Presentation. Barnsley G, ..., **Watson OJ**. Global impact of the first year of COVID-19 vaccination: a mathematical modelling study. 16th Vaccine Congress. 2022

Oral Presentation. Watson OJ. Novel uses of satellite imagery for pandemic preparedness and mortality estimation in humanitarian settings. Innovation Endeavours Invited Pitch. 2022

Poster Presentation. Paschalidis A, **Watson OJ** et al. Complexity of Infection Estimation with Allele Frequencies. ASTMH. 2021

Oral Presentation. Okell L, **Watson OJ** et al. Multiple first line therapies versus reducing overprescription of antimalarials to slow antimalarial resistance. ASTMH. 2018

Poster Presentation. **Watson OJ** et al. The impact of seasonal variation in the detection of clinically relevant plasmodium falciparum hrp2 gene deletions: a modelling study. ASTMH. 2018

Poster Presentation. **Watson OJ** et al. Evaluating the performance of malaria genomics for inferring changes in transmission intensity using transmission modelling. Genomic Epidemiology of Malaria. 2018

Poster Presentation. Cremin I, **Watson OJ** et al. An infectious way to teach students about outbreaks. Epidemics 6. 2017

Oral Presentation. **Watson OJ** et al. rdhs: an R package to interact with the demographic and health surveys (DHS) program data sets. ASTMH. 2017

Oral Presentation. **Watson OJ** et al. Characterizing the potential bias within genomic tools for inferring changes in plasmodium falciparum transmission intensities. ASTMH. 2017

Poster Presentation. **Watson OJ** et al. Modelling the epidemiological implications and explanations of discordances between microscopy and rapid diagnostic tests. Future of Malaria Research Symposium. 2017

Oral Presentation. **Watson OJ** et al. Drivers of the spread of “diagnostic resistant” P. falciparum malaria: a model-based evaluation of the spread of pfhrp2 gene deletions in Africa. ASTMH. 2016

Invited Oral Presentation. **Watson OJ** et al. Modelling the drivers of Plasmodium falciparum hrp2 deletions. WHO organised panel meeting on “Plasmodium falciparum hrp2/3 gene deletions: update, implications and response” at ASTMH. 2016

TEACHING EXPERIENCE

PhD Supervision:

Nisreen Shehada. Imperial College London. Vaccine preventable disease models to support humanitarian operations. Primary supervisor; secondary supervisor Aula Abbara. Started January 2026.	2026 -
Cosmo Santoni. Imperial College London. Deep learning surrogates for infectious disease models. Primary supervisor; secondary supervisor Thomas Churcher. Started January 2025.	2025 -
Venitha Bernard. Temple University. Statistical methods for inferring drug resistance phenotypes for antimalarial resistance genotypes. Supervisory committee member; primary supervisor Maciej Boni.	Current
Isabela Gerdes Gyuricza. University of North Carolina at Chapel Hill. Global determinants for the evolution and selection of pfhrp2/3 gene deletions in malaria. Supervisory committee member; primary supervisor Jonathan Parr.	Current
Cecile Meier-Scherling. Brown University. Comparison of strength of selection for P. falciparum artemisinin resistance-associated mutations between southeast Asia and Uganda.	2022 -
Ruth McCabe. University of Oxford. Epidemiological analysis and modelling of COVID-19 and other high consequence infectious disease transmission risks.	Completed 2024

Masters Supervision:

MSc Epidemiology. Amber Dunmore. Imperial College London. Systematic Review of Sexual Commitment of Artemisinin Resistant Malaria to Inform Mathematical Models.	2025
MPH. Shuyi Liu. Imperial College London. Systematic Review of the Impact of COVID-19 Testing Strategies on Public Health Outcomes to Inform WHO COVID-19 Testing Guidance.	2025
MPH. Nisreen Shehada. Imperial College London. Systematic Review on Studies of Interventions to Address the Infectious Disease Health Needs of Populations in Humanitarian Settings and Inform Vaccine Preventable Disease Models for Gaza.	2025
MPH. Roshwini Muthia. Imperial College London. Cost-effectiveness of Expanding Seasonal COVID-19 Vaccination for All Age Groups: Impact on Productivity Loss Prevention in the UK.	2025
MPH. Mariam Diefallah. Imperial College London. Systematic Review on Studies of Interventions to Address the Non-Communicable Disease and Sexual and Reproductive Health Needs of Populations in Humanitarian Settings.	2025
MSc Epidemiology. Hallie Benjamin. LSHTM. Return on Investment of COVID-19 Vaccines.	2024
MRes Biomedical Research, Epidemiology, Evaluation and Control of Infectious. Chaokun Hong. Modelling the impact of drug resistance on the selection of resistance to front line antimalarial treatment.	2024
MSc Epidemiology. Marco Chan Yan Chit. Modelling the impact of drug resistance on the selection of resistance to front line antimalarial treatment.	2024

Undergraduate Supervision:

Rebecca Kirby. Brown University. Examining the Early Distribution of the Artemisinin-Resistant Plasmodium falciparum kelch13 R561H Mutation in Areas of Higher Transmission in Rwanda. Published in Open Forum Infectious Diseases, Volume 10, Issue 4, April 2023, ofad149.	2020 - 2021
Aris Paschalidis. Brown University. Direct estimation of COI from whole genome sequence data. Published in PLoS Computational Biology. 2023;19(6):e1010247.	2020 - 2022

Teaching:

Module co-lead on the five-week Outbreaks module. MSc Epidemiology, Imperial College London.	2025/26 -
“Planning and preparing outbreak research”. Three-hour lecture and practical in Emerging and Neglected Tropical Diseases, MSc Epidemiology, Imperial College London.	2025/26 -
Master’s in AI in Science, African Institute for Mathematical Sciences, Cape Town. One week of teaching	2024/25 -

for approximately 30 students, including lectures, assessment design and marking.

Co-organised the five-day Nairobi modelling hackathon with CEMA, University of Nairobi, and Kenya's Ministry of Health. Group facilitation and GitHub training. January 2025. 2025

DIDE internal training: Introduction to deep learning frameworks with Python and PyTorch. Three-hour session with Cosmo Santoni. 2025

"Outbreaks in Emergency Crises". One-hour lecture in the Outbreaks module, MSc Epidemiology, Imperial College London. 2024/25 -

Introduction to R. INES-Ruhengeri, Rwanda. Training in reproducible epidemiological analysis. 2024

Module co-lead on the 16-week Health Protection module. Imperial College London GMPH. 2024/25 -

Weekly distance learning for 5 Somali researchers at SIMAD university for mortality estimation in R. 2023

Demonstrator on 2-day practical on developing a model to answer a public health control policy question. Imperial College London. MSc Epidemiology. 2016 - 2021

Introduction to R and statistical modelling. 3 day introductory course for computational biology rotation students. Brown University. 2020

Organised teaching material and demonstration assistance on Imperial College London 2-week short course on infectious disease modelling. 2016 - 2018

Prepared computer-based practical exercise on outbreak reconstruction. Imperial College London. MSc Epidemiology. The practical exercise and its administration were subsequently published as a pedagogical exercise ("An infectious way to teach students about outbreaks". *Epidemics*. 23, 42–48 (2018)) 2016

Demonstrator on the use of multi locus sequence typing databases in epidemiology. Imperial College London. MSc Epidemiology 2016

Demonstrator on introductory excel refresher. Imperial College London. MSc Epidemiology 2016

Professional short-course teaching. Developed and delivered a full one-day lecture and practical on stochasticity in infectious disease modelling for the Mathematical Modelling for the Control of Infectious Diseases short course, Imperial College London. Current teaching recorded in January 2026.

Research staff supervision: Timothy Hitge (PDRA, deep-learning surrogates, November 2025 - present); Arran Hamlet (PDRA, JI-RISE, March - August 2025); Mantra Susskind (research software engineer, JI-RISE, March 2025 - present); Paula Christen (visiting researcher/consultant, Wellcome Ideathon and JI-RISE, April 2024 - present).

For my teaching demonstrations, I was nominated for a Student Academic Choice Awards (SACA) Teaching Award (2019).

I am passionate about improving my supervision approach and mentorship and have undertaken both external training on career development offered by Stanford University and provide evaluation forms from all students and researchers I supervise and manage (which have all been scored as excellent (6 out of 6), N = 5). This was an initiative I chose to start, both to understand how I can improve as I develop seniority and more frequently supervise other researchers but also in response to expanding my research supervision and support across time zones and country borders.

SOFTWARE DEVELOPMENT

I develop and maintain R packages and reproducible research compendia for infectious disease modelling, demographic data access and malaria genetics. Selected software and research compendia are listed below.

malariaAtlas. Co-author of the R interface to open-access Malaria Atlas Project parasite-rate surveys, mosquito occurrence data and raster surfaces. <https://github.com/malaria-atlas-project/malariaAtlas>

hrpup. Reproducible research compendium for hrp2 deletion risk mapping. <https://github.com/OJWatson/hrpup>

arms. Reproducible research compendium for antimalarial resistance risk mapping. <https://github.com/OJWatson/arms>

hmmibdr. R implementation of the hmmIBD method for estimating identity by descent in malaria infections. <https://github.com/OJWatson/hmmibdr>

rdhs. API Client and Data Munging for the Demographic and Health Survey Data. Cited in 27 publications. <https://github.com/ropensci/rdhs>

squire. SEIR transmission model of COVID-19 and model fitting tools. Cited in 14 publications. <https://github.com/mrc-ide/squire>

nimue. Extension to the squire package that allows vaccination to be included. Cited in 6 publications. <https://github.com/mrc-ide/nimue>

McCOILR. R implementation of The REAL McCOIL method for estimating complexity of infection in malaria infections. Cited in 3 publications. <https://github.com/OJWatson/McCOILR>

hrp2malaria. Individual based model of Imperial College malaria model for modelling hrp23 deletions. Cited in 3 publications. <https://github.com/OJWatson/hrp2malaria>

safir. Individual based rewrite of the squire and nimue COVID-19 models. Cited in 2 publications. <https://github.com/mrc-ide/safir>

outbreakteachR. Accompanying R package for assisting in the analysis and execution of a simulated outbreak amongst students. Cited in 1 publication. <https://github.com/mrc-ide/outbreakteachR>

magenta. Spatial, individual based model of Imperial College malaria model for modelling parasite genetics. Cited in 1 publication. <https://github.com/OJWatson/magenta>

PROFESSIONAL MEMBERSHIP AND SERVICE

Guest Editor. BMJ Global Health topic collection on Population Mortality in Fragile and Conflict Settings. (2025).

Co-organiser. IDE Seminar Series, Department of Infectious Disease Epidemiology, Imperial College London. (2025 - present).

Co-lead. MRC Centre for Global Infectious Disease Analysis Equity and Inclusion in Global Health Working Group. (2024 - present).

Co-lead. Development of the School of Public Health Workload Allocation Model. Imperial College London. (2025 - present).

Lead reviewer. School of Public Health teaching-quality evaluation, reviewing five years of feedback across programmes to inform teaching strategy. (2024 - January 2025).

Member. Faculty of Medicine AI Working Group; co-organised the one-day collaborative AI in Medicine workshop with 101 applicants. Imperial College London. (2024 - present; workshop 18 February 2025).

Fellowship Representative. Eric and Wendy Schmidt AI in Science Fellowship, I-X, Imperial College London. (2023 - present).

Member. American Society of Tropical Medicine and Hygiene. (2024 - present).

Internal fellowship reviewer and mentor. Schmidt Science Fellowship, Imperial College London. Internal assessment and coaching of a successful nominee. (2025).

PhD Representative. Department of Infectious Disease Epidemiology, Imperial College London. (2018 - 2019).

Reviewer. Nature, PNAS, Nature Communications, The Lancet, PLoS Medicine, eLife, PLoS Computational Biology, Genome Biology and Evolution, PLoS Global Public Health, American Journal of Tropical Medicine & Hygiene, Evolutionary Applications, Malaria Journal, Conflict and Health, Royal Society Interface, Lancet Regional Health

Grant Reviewer. Wellcome Trust, NIH, Horizon Europe, Netherlands Organisation for Health Research and Development, Imperial College London (Internal Fellowship Rounds)

Shadow Interview Panels. Imperial College London (2023)

Interview Panels. Imperial College London (2021, '22, '23), LSHTM (2022, '23)

Examiner. PhD Advisory Committee (Temple, UNC at Chapel Hill), Imperial PhD (Early Stage Assessment), Imperial MRes (2024), Imperial GMPH (2024), Imperial MSc (2024, 2021), Imperial MPH (2024, 2021), Oxford MSc (2021, '23); external marker, Oxford MSc Modelling for Global Health (2022 - present)

Member. Afya Consortium: US CDC funded consortium of academic and civil society organisations, working equitably to strengthen research capacity, improve data equity and conduct research in humanitarian settings.

Member. WHO pfhrp23 technical panel: Supporting the update of the WHO response plan to pfhrp23 deletions, providing expertise and modelling to support malaria diagnostic policy.

Member. WHO MFT technical panel: Provided expertise and modelling informing the 2024 guidance on multiple first-line therapies for malaria, supporting malaria treatment policy.

Member. ROpenSci: Community of software developers building open source and reproducible tools using the R programming language. I have contributed R packages and code reviews for data access and survey tools.

Member. VacSafeWG: Schmidt Futures funded group for furthering vaccine safety and confidence-building in Africa. Contributed to policy briefs on vaccines, data sources and early warning systems for supranational agencies. (2022 - 2024).

PUBLIC ENGAGEMENT AND OUTREACH

School work experience. Hosted two high-school students for a one-week research placement at Imperial College London. (2024).

Various media appearances including TV (Sky, BBC, Euronews, TV Globo), Radio (BBC, RTS, Sky) and News (FT, Times, Guardian, Washington Post, NY Times, Economist - cover story) related to my COVID-19 work. Additionally, I provide to a number of fact-checking agencies (AFP Fact Check, German Press Agency dpa).

Science in Context Imperial College London Videos to discuss findings from our research.

Facebook Live Community Outreach with Sudanese Health Influencers. Conducted a series of live interviews and AMA with an Arabic interpreter to explain the impacts of our research in Khartoum (2020).

RISE Residential Summit. Interactive classroom simulation and pandemic tabletop exercise showcasing the role of infectious disease modelling in policy making. Delivered to the 100 winners of the RISE Global Scholarship program (2023).

Central London Data Science. Organiser of a meetup group teaching fundamentals of data science to over 3000 members across industry, academia and members of the public. (Founded in 2016).

Brilliant Club Course Author. Co-authored an 8-week Key Stage 2 UK (ages 7-11) curriculum entitled "Disease Detectives" (2018).